

Automated Fishtank Monitoring and Care System Via Embedded System

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Abstract—This project presents a low-cost embedded system that automates fish tank monitoring and feeding using temperature and turbidity sensing, a servo feeder, and a simple user interface. The prototype operates on a minimal and simple software stack, providing an accessible, reliable drop-in solution for beginner aquarium owners.

I. INTRODUCTION

Many fish owners struggle to maintain stable tank conditions because affordable, all-in-one automation systems are rare. Existing solutions are often costly, relying on continuous internet connectivity and complex software, or requiring an expensive and time-consuming setup. This gap in the market motivates the development of a simple, low-power, low-maintenance drop-in system that automates monitoring and basic aquarium care while remaining affordable and accessible.

The objective of this project is to build an embedded system that provides real-time temperature and turbidity monitoring, an intuitive visual interface, programmable feeding intervals, a buzzer that alerts users when the tank becomes overly cloudy, and weekly reminders to manually check pH levels. Rather than relying on expensive sensors or mobile connectivity, the design prioritizes simplicity and cost-effectiveness while delivering essential functionality.

Previous literature highlights a trend toward IoT-based aquarium monitoring. Rajesh et al. [1] describe a smart tank system that uses an ESP32 to send alerts and regulate conditions through automated water flow, demonstrating effective remote oversight but requiring network infrastructure and email server integration. Rey [2] presents FishTankv2, which uses machine learning to predict maintenance needs and relies on mobile applications for control, offering sophisticated capabilities but adding complexity and cost. Another study [3] outlines a real-time IoT aquarium management system with mobile monitoring, temperature and pH sensors, and automated feeding operations. While these systems provide advanced functionality, they all depend on wireless connectivity, cloud-based communication, and, in some cases, expensive or high-maintenance sensors such as pH probes.

In contrast, the system described in this report avoids Wi-Fi, machine learning, and costly sensing components. Eliminating pH sensors removes the need for frequent calibration and specialty materials, replacing them with simple auditory tank maintenance reminders. The absence of wireless communication also removes security concerns and firmware maintenance burdens. The project's core innovation lies in

providing a simplified, fully self-contained appliance that hangs on the side of any tank, requires only a simple power source, and operates without external dependencies. Despite the reduced complexity, the system aims to retain the most essential features required for effective aquarium care.

II. SYSTEM DESIGN

The system architecture was initially outlined through hardware and software block diagrams that guided component selection and modular development. The embedded platform centers on a TI-TM4C series microcontroller chosen for its familiarity to the team and compatibility with an existing custom RTOS. This RTOS supports modular integration of sensors while providing useful debugging capabilities and structured task management that would be more difficult to achieve using simple loops and interrupts.

A. Hardware Architecture

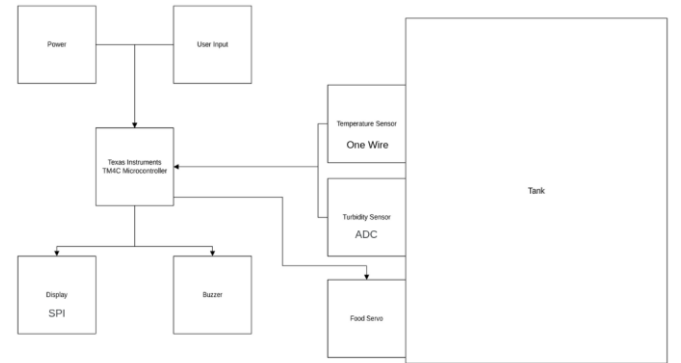


Fig. 1. Hardware Block Diagram

The hardware subsystem consists of several major components: the microcontroller unit (MCU), environmental sensors, actuation hardware, user interface elements, and power management circuitry. The TM4C microcontroller serves as the central processing unit responsible for interpreting sensor data, driving actuators, and coordinating periodic tasks. Its wide support for analog and digital peripherals including ADC channels, timers, PWM generators, and GPIO hubs makes it an appropriate choice for a multi-sensor embedded project.

Environmental monitoring is achieved using two primary sensors. Temperature is measured through a digital DS18B20

with enhanced precision at the cost of a more complex one wire protocol. Turbidity is measured using the SEN0189 analog sensor, which outputs a voltage proportional to water clarity. Because turbidity sensors are sensitive to orientation and submersion depth, the design incorporates a waterproof or floating housing to preserve signal consistency while protecting the electronics.

Actuation is handled by an SG90 continuous-rotation micro servo used to dispense food from the feeder mechanism. The servo is driven by PWM signals generated by the MCU's hardware timers. Its rotational speed and duration are tuned to deliver a controlled amount of food per cycle, preventing overfeeding. A buzzer system with an active buzzer alerts users when turbidity levels exceed a predefined threshold, serving as an indicator that the tank requires cleaning.

User interaction is supported by an EA LED55X46-A display and pushbutton input system. The LCD display shows real-time measurements, feeding intervals, and data graphing. The pushbuttons enable users to set the feeding time and navigate interface states. The peripherals are powered through a standard power supply, with efforts made to keep current draw under 500 mA to maintain compatibility with low-cost power adapters. The microcontroller was powered through the computer and programming interface to minimize the risk of hardware damage during development and to accommodate time constraints as the demonstration deadline approached.

A significant portion of the hardware design also involves 3D-printed components custom-fit for the aquarium environment. These include hooks that secure the system to the tank wall, brackets for sensor placement, and a controlled dispenser assembly for the servo-based feeder. Mechanical design considerations include water exposure, load distribution, and easy detachment for cleaning or maintenance.

B. Software Architecture



Fig. 2. Software Block Diagram

The software subsystem relies on a custom real-time operating system tailored to the TM4C platform. This RTOS enables modular programming by separating major system tasks into discrete threads or scheduled events. This approach prevents blocking operations such as sensor sampling or

displays updates from interfering with feeding logic or timing routines. Compared to a traditional loop-and-interrupt architecture, RTOS offers substantially improved system clarity, maintainability, and debugging visibility throughout the development process.

Sensor routines are executed at periodic intervals using RTOS timers. Temperature is acquired through a custom developed one wire driver, and turbidity readings are acquired through ADC sampling and processed using a sliding average filter to yield meaningful physical values. The turbidity processing module compares clarity readings against user-defined thresholds to determine whether the buzzer should activate.

Feeding logic is managed through the main control loop task and a timed task to manage servo rotation. The control task is responsible for tracking elapsed time using a real-time clock reference, and the servo task ensures that the servo rotates around one slot in addition to performing the set number of feedings.

The control task also updates the display module with current sensor values, feeding timers, and warning indicators. To maintain stability, the display is refreshed at a moderate frequency to preserve readability while reducing unnecessary computational load. Button inputs are debounced and processed through additional tasks that post events to a shared memory interface, ensuring responsive control without blocking other tasks.

C. Physical Integration

The final stage of system design focuses on integrating the hardware onto a single prototyping board. The entire unit attaches on the tank edge using 3D-printed mounts that both stabilize the electronics and position the sensor float at the correct depth. Cable routing is planned to avoid water exposure and provide strain relief. The goal is to produce a device that is not only functional but also durable and user-friendly in a real aquarium environment.

III. RESULTS AND DISCUSSION

At the demonstration stage, several subsystems have been successfully completed and validated. The servo feeder mechanism operates through a PWM signal that consistently dispenses measured amounts of food. The temperature sensor reliably reports real-time tank temperature on the LCD display, and the turbidity sensor produces accurate analog readings that are calibrated against a threshold value for water clarity.

Updated hardware and software block diagrams reflect adjustments made after subsystem validation, such as one wire over other protocols to reflect use of the DS18B20. Additional updates include attaching 3D-printed hooks to mount the device onto the tank, integrating the buzzer to signal turbidity thresholds, designing a floating waterproof enclosure for the turbidity sensor, consolidating wiring into a prototyping board, and implementing RTC timer logic for the feeder.

In its completed form, the system provides a fully integrated system, continuously monitoring temperature and turbidity, operating the feeder at user-configured intervals, activating the buzzer when water quality degrades, and displaying essential information for quick reference. The device meets key performance benchmarks, including accurate temperature measurement, consistent turbidity readings, reliable feeding with at least 90% success, prompt buzzer activation, clear visual output, intuitive button interaction, efficient power consumption, and stable long-duration operation.

IV. CONCLUSION

Overall, the project demonstrates how simple embedded systems can provide meaningful automation for fish tank owners without relying on costly sensors, cloud-based systems, or machine learning. By focusing on simplicity, accessibility, and cost reduction, the design offers a practical entry-level automation tool for new fish keepers. The prototype provides real-time environmental data, turbidity-based cleanliness alerts, scheduled feeding, and a user-friendly interface. The completed system shows strong potential for expansion in the future, including improvements in user interface design, sensor quality, and environmental control.

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